

Loss-Aware Synthetic Data Generation: Estimating Utility Loss and Preventing It

Ali Saleh Mohammadabad
ali.saleh4@studio.unibo.it
Matriculation Number: 0001153960

Abstract

Synthetic data generation is increasingly used as a privacy-preserving technique for sharing and analysing data without directly exposing sensitive real records. However, synthetic data may lose important information from the original dataset, such as statistical distributions, correlations between features, or predictive patterns required for downstream machine learning tasks. This project focuses on *loss-aware synthetic data generation*, where the quality of synthetic data is evaluated not only in terms of privacy protection, but also in terms of the utility loss introduced when real data is replaced by synthetic data.

The goal of the project is to design a small experimental framework for measuring the privacy–utility trade-off in synthetic tabular data. The project will use the UCI Adult / Census Income dataset as a case study and compare real and synthetic data using distributional similarity, correlation preservation, and downstream classification performance. Privacy-related risks will also be considered through distance-based or inference-risk indicators. If feasible, the project will explore a simple loss-aware strategy to reduce utility loss by introducing penalty terms related to distribution mismatch, correlation distortion, or performance degradation.

Motivation

Synthetic data can support responsible AI development by enabling experimentation, data sharing, and model training while reducing the direct exposure of personal or sensitive information. This is particularly relevant when real datasets contain demographic, socioeconomic, medical, or behavioural information that cannot be freely distributed.

However, privacy alone is not sufficient. If synthetic data protects privacy but does not preserve the useful structure of the original data, models trained or evaluated on it may become unreliable. For example, synthetic data may distort feature distributions, remove important correlations, or reduce predictive performance in downstream tasks. This creates both a technical and ethical problem: the generated data may appear safe, but it may lead to misleading analysis or poor decision-making.

For this reason, the project studies synthetic data generation from a Trustworthy AI perspective, focusing on the balance between privacy protection and utility preservation.

Dataset

The project will use the *UCI Adult / Census Income* dataset as the main experimental dataset. This is a public tabular dataset commonly used for binary classification. The prediction task is to determine whether an individual’s annual income exceeds 50K based on demographic and employment-related attributes.

This dataset is suitable for the project because it contains both numerical and categorical features, has a clear target variable, and allows the evaluation of synthetic data utility through downstream machine learning performance. In addition, the presence of demographic attributes makes it relevant for ethical discussion around privacy, responsible data use, and possible bias in data-driven systems.

Goals

The main goal of this project is to investigate how much useful information is lost during synthetic data generation and how this loss can be measured and possibly reduced.

More specifically, the project aims to:

1. define utility loss in synthetic tabular data;
2. generate synthetic versions of a real tabular dataset;
3. compare real and synthetic datasets using quantitative metrics;
4. evaluate whether synthetic data preserves feature distributions and correlations;
5. measure the impact of synthetic data on downstream machine learning performance;
6. consider privacy-related risks associated with generated records;
7. discuss the ethical implications of the privacy–utility trade-off;
8. optionally explore a simple loss-aware strategy for reducing utility loss.

Methodology

The project will start by preprocessing the selected dataset and defining a baseline classification task. Then, one or more synthetic data generation methods will be used to create synthetic versions of the original dataset.

The evaluation will consider two main perspectives. First, **utility preservation** will be analysed by comparing real and synthetic data in terms of distributional similarity, correlation preservation, and downstream classification performance. Possible metrics include differences in feature distributions, correlation matrix distance, and the discrepancy between the performance of a model trained on real data and a model trained on synthetic data.

Second, **privacy protection** will be considered through simple privacy-risk indicators. These may include distance-based comparisons between real and synthetic records or inference-risk indicators. The objective is not to claim full privacy guarantees, but to evaluate whether the generated data exposes patterns that may create privacy concerns.

If time permits, the project will also explore a simple loss-aware strategy. This could involve adding explicit penalty terms or selection criteria that discourage synthetic datasets with high distribution mismatch, high correlation distortion, or poor downstream performance.

Expected Deliverables

The expected deliverables are:

1. an initial report describing the project idea, motivation, goals, and expected outcomes;
2. a GitLab repository containing the code, instructions, and experimental material;

3. a reproducible evaluation pipeline for comparing real and synthetic datasets;
4. experimental results showing the trade-off between utility preservation and privacy protection;
5. a final report discussing the methodology, results, ethical implications, limitations, and possible future improvements;
6. if feasible, a prototype loss-aware approach for reducing utility loss during synthetic data generation.

Expected Outcome

The expected outcome is a computational framework for evaluating synthetic data generation in terms of both privacy and utility. The project should provide a clearer understanding of what is lost when synthetic data replaces real data, how this loss can be measured, and how it may be reduced. The final discussion will connect the technical results to broader ethical issues in AI, especially privacy-preserving AI, responsible data use, robustness, and trustworthy machine learning.